Lecture 25

Posterior predictive checks

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(Edited from the lecture notes by Brian J. Reich, NC State)

Posterior predictive checks

- After comparing a few models, we settle on the one that seems to fit the best.
- Given this best model, we then verify if it is adequate.
- The usual residual checks are appropriate here: e.g., qq-plots.
- A uniquely Bayesian diagnostic is the posterior predictive check.
- This leads to the Bayesian p-value.

Posterior predictive distributions

- Before discussing posterior predictive checks, let us review Bayesian prediction in general.
- The plug-in approach would fix the parameters θ at the posterior mean $\hat{\theta}$ and then predict $Y_{new} \sim f(y|\hat{\theta})$.
- This suppresses uncertainty in θ.
- We would like to propagate this uncertainty through to the predictions.

Posterior predictive distributions (PPD)

- We really want the PPD

$$f(Y_{new}|\mathbf{Y}) = \int f(Y_{new}, \theta|\mathbf{Y})d\theta = \int f(Y_{new}|\theta)f(\theta|\mathbf{Y})d\theta.$$

- MCMC easily produces draws from this distribution.

- To make S draws from the PPD, for each of the S MCMC samples of θ, we draw a Y_{new}.

- This gives draws from the PPD and clearly accounts for uncertainty in θ.

Posterior predictive checks

- Posterior predictive checks sample many datasets from the PPD with the identical design (same n, same $\mathbf{X}$) as the original data set.
- We then define a statistic describing the dataset, e.g.,

$$d(\mathbf{Y}) = \max\{Y_1, ..., Y_n\}.$$

- Denote the statistic for the original data set as d_0 and the statistic from simulated data set number s as d_s.
- If the model is correct, then d_0 should fall in the middle of the $d_1, ..., d_S$.

Posterior predictive checks

- A measure of how extreme the observed data is relative to this sampling distribution is the Bayesian *p*-value.

$$p = \frac{1}{S}\sum_{s=1}^{S}\mathbb{I}(d_s > d_0).$$

- If p is near zero or one the model doesn't fit.

- This is repeated for several d to give a comprehensive evaluation of model fit.

Thank you!